

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/D_DEB_SAAWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3303, C2 = 3666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (K,           Uniform,                0.00001,   2);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, seeds)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kap.1. kapA.1.

Result_mode	0.002548330	164.58600	0.76404300	0.20257900	0.92967900	0.51047500
2.5%	0.000562278	148.41700	0.62825400	0.15669500	0.87848275	0.40102300
97.5%	0.004768873	188.48200	0.92934200	0.22653900	0.95252900	0.60512100
Min	0.000471029	104.83800	0.52459900	0.13109000	0.12674200	0.21707400
Max	0.320981000	212.54200	1.03398000	0.25731300	0.95840100	0.68186700
Result_mean	0.002460287	166.59510	0.77585313	0.19150764	0.92139589	0.50505525
Result_sd	0.002986915	11.03929	0.07157585	0.01804138	0.02016055	0.04960192
Ehp.1. E_coc.1. rOM_ClxHorse.1. K.1. dv.1.						
Result_mode	95.10720	30.27920	0.007926100	0.01745650	3.9496000	
2.5%	65.85159	22.07560	0.005854410	0.00437267	3.2525260	
97.5%	132.13100	37.09200	0.011446800	0.03554690	5.8477800	
Min	54.30420	17.58680	0.000281582	0.00369115	0.5220400	
Max	160.76000	41.42600	0.289479000	1.59821000	6.5286900	
Result_mean	95.21560	29.07275	0.008628415	0.02067126	4.3987272	
Result_sd	18.24274	3.76644	0.002209851	0.01475497	0.6498835	
wE.1. Sigma_W.1.						
Result_mode	1.341750e-09	0.12757000				
2.5%	1.244240e-09	0.11794315				
97.5%	5.859156e-06	0.15030300				
Min	1.139880e-09	0.10575500				
Max	1.091100e-05	5.52230000				
Result_mean	6.119552e-07	0.13323798				
Result_sd	1.471612e-06	0.05254083				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -36.4097
 BIC = 140.3076

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.07	1.17
E_coc.1.	1.07	1.13
Ehp.1.	1.35	1.94
Fm.1.	1.57	2.86
K.1.	1.55	3.02
kap.1.	1.16	1.48
kapA.1.	1.14	1.41
pAm.1.	1.19	1.58
r_pAm_pM.1.	1.08	1.24

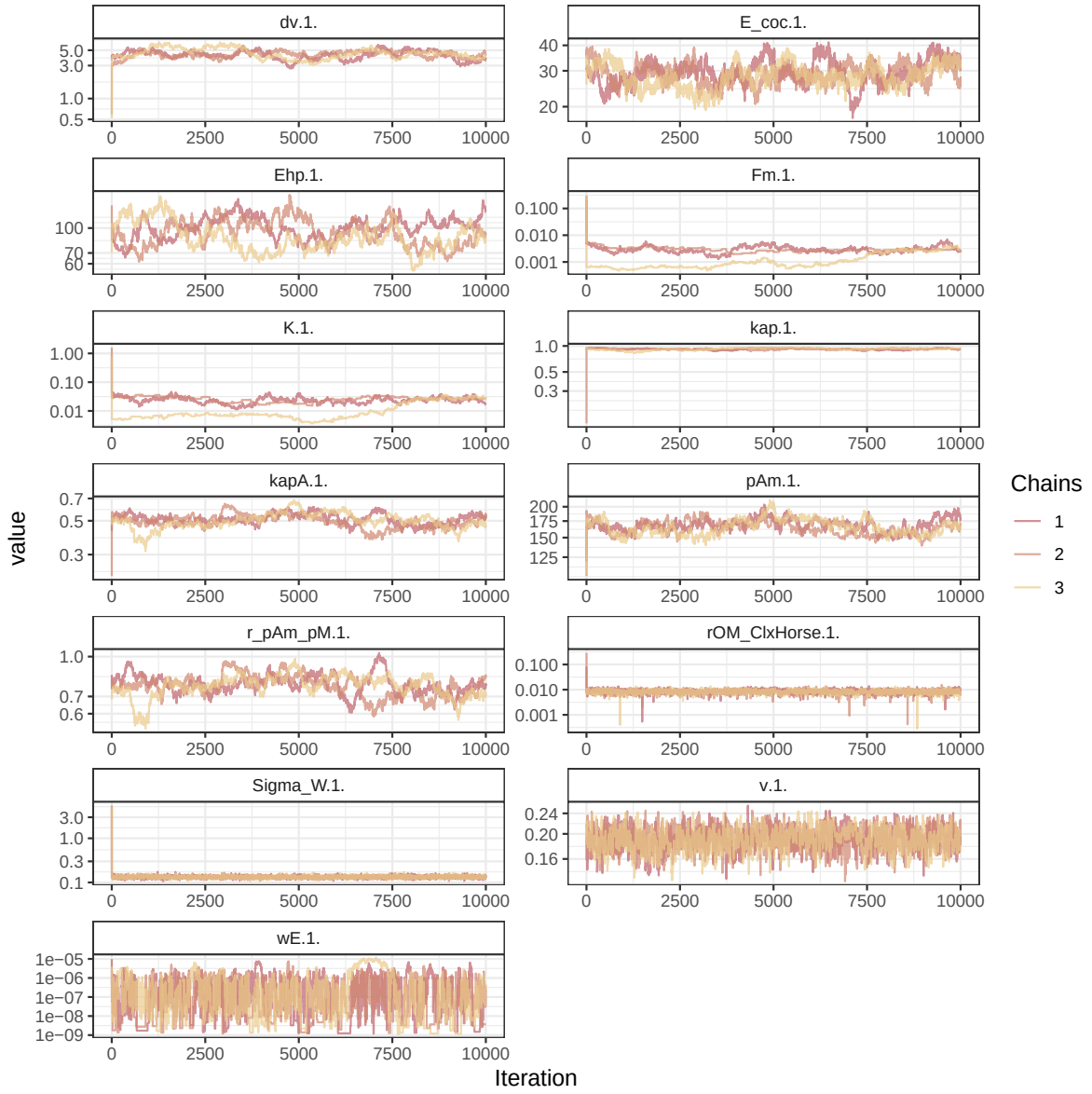


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

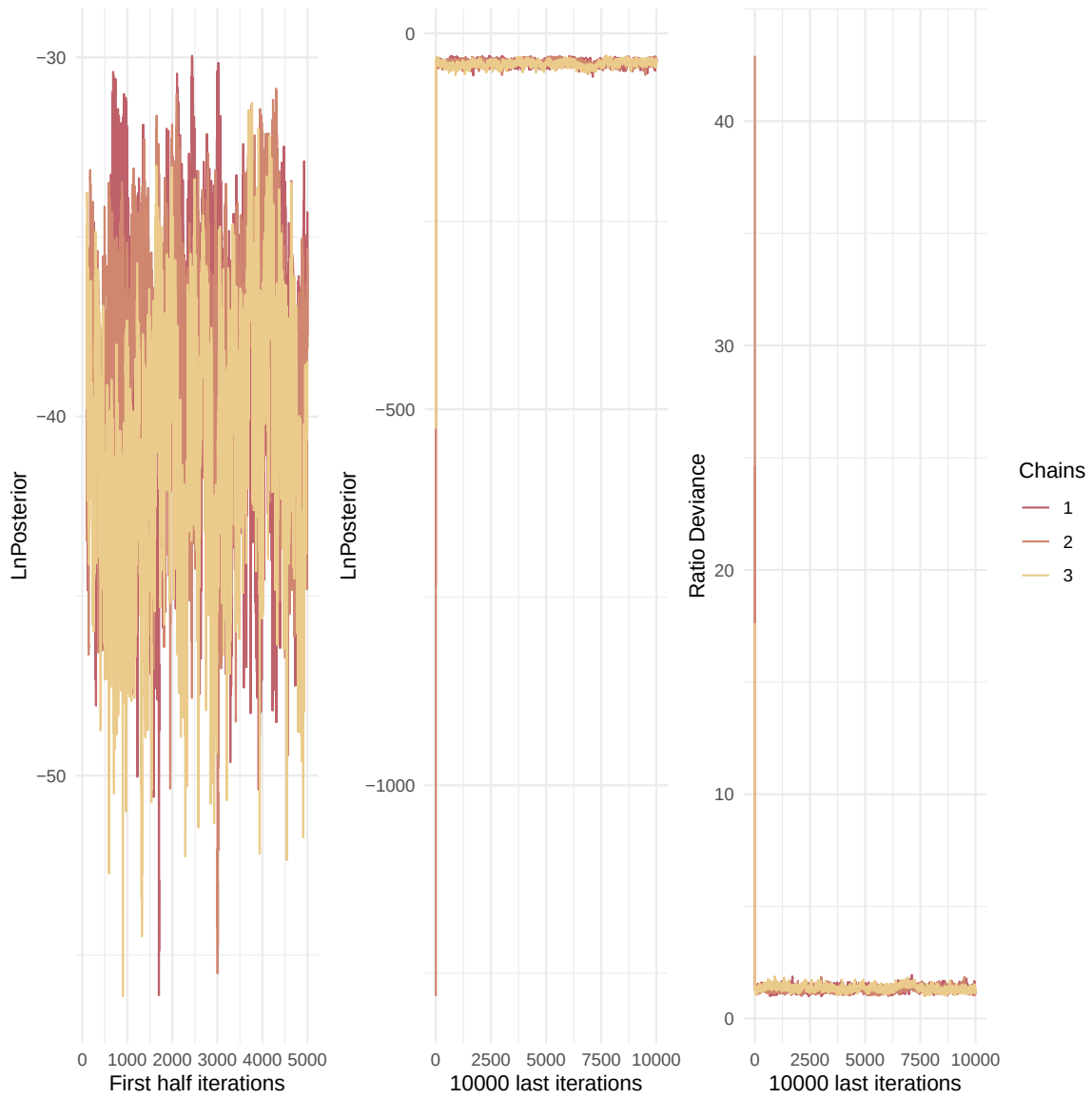


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.01
v.1.	1.00	1.01
wE.1.	1.30	2.29

Multivariate psrf

1.55

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

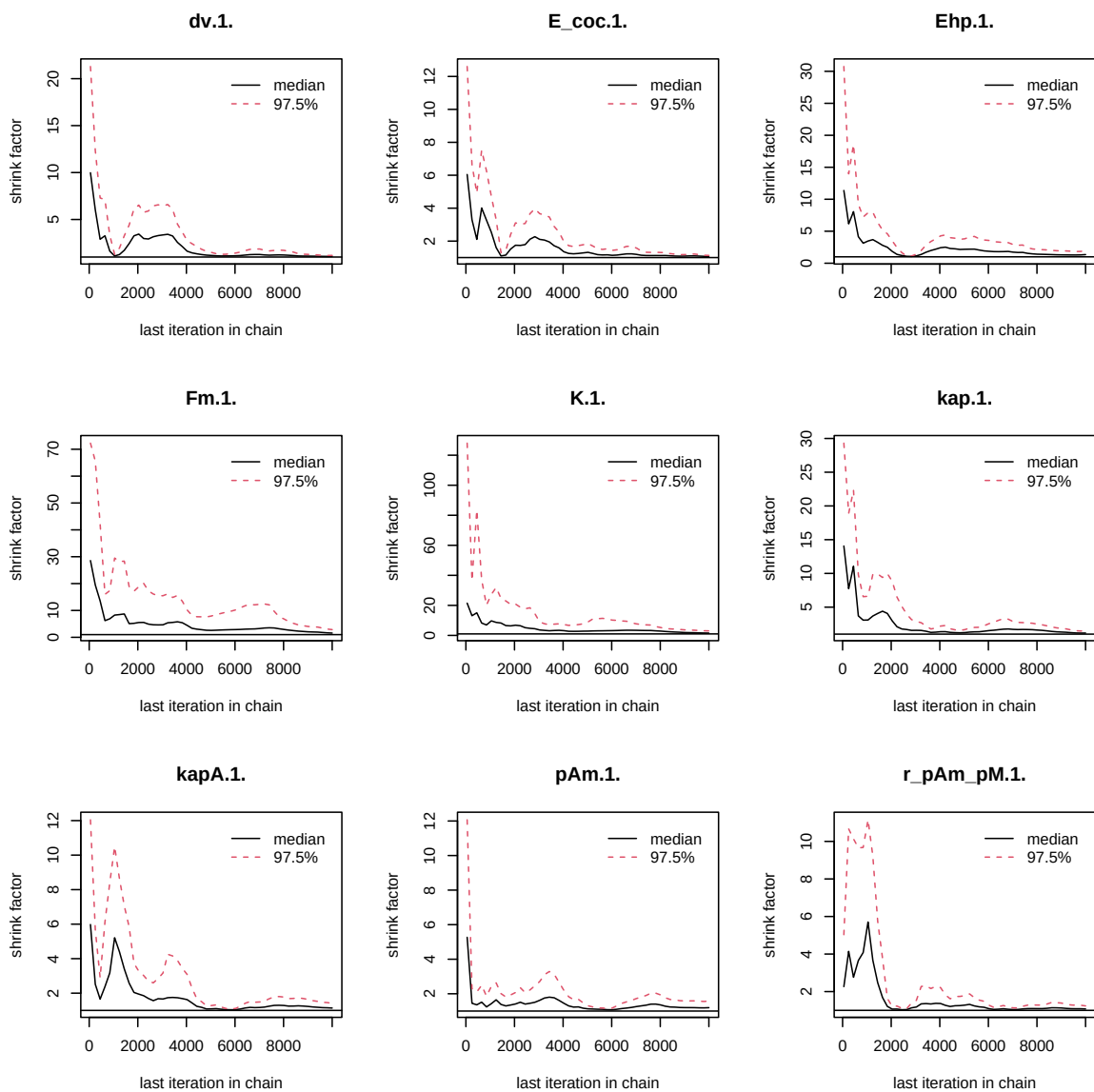


Figure 3: Chains convergence.

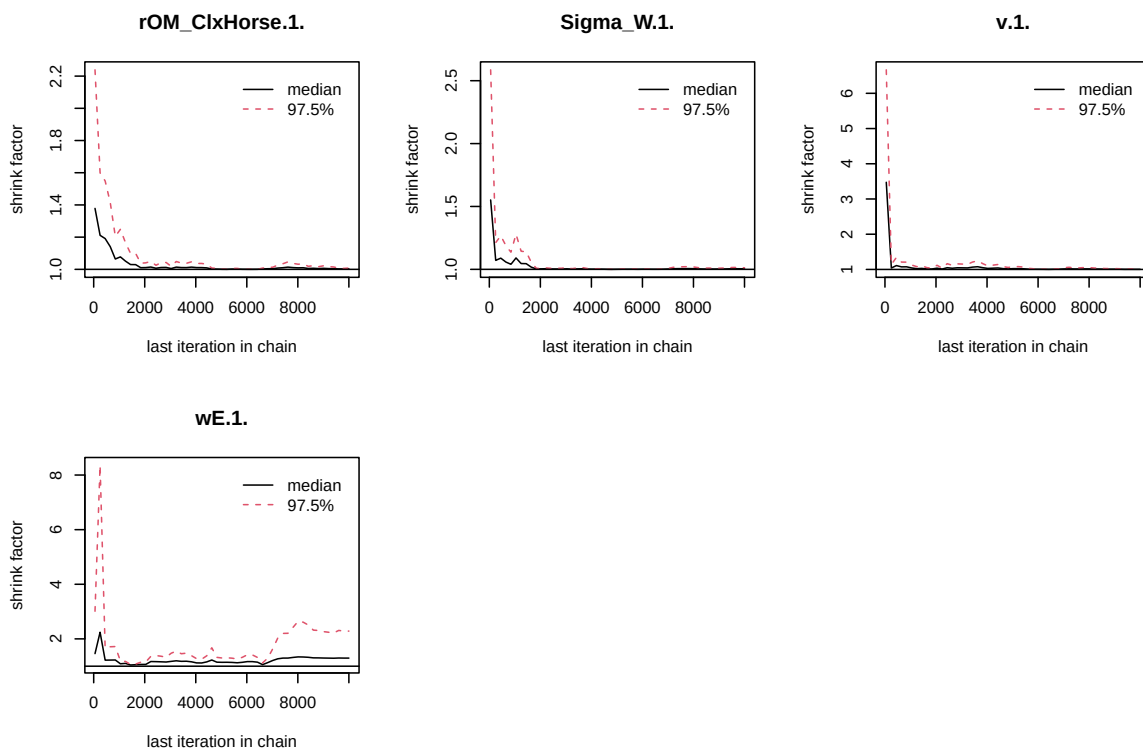


Figure 4: Chains convergence.

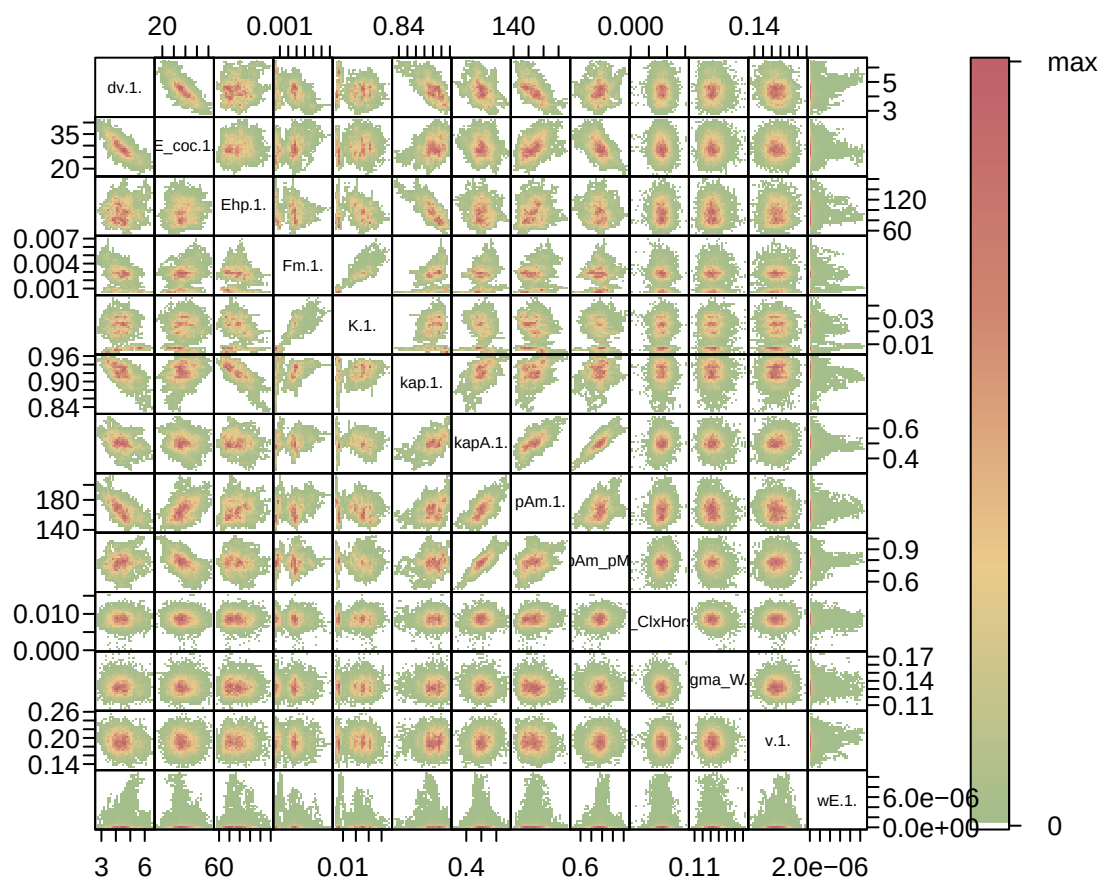


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

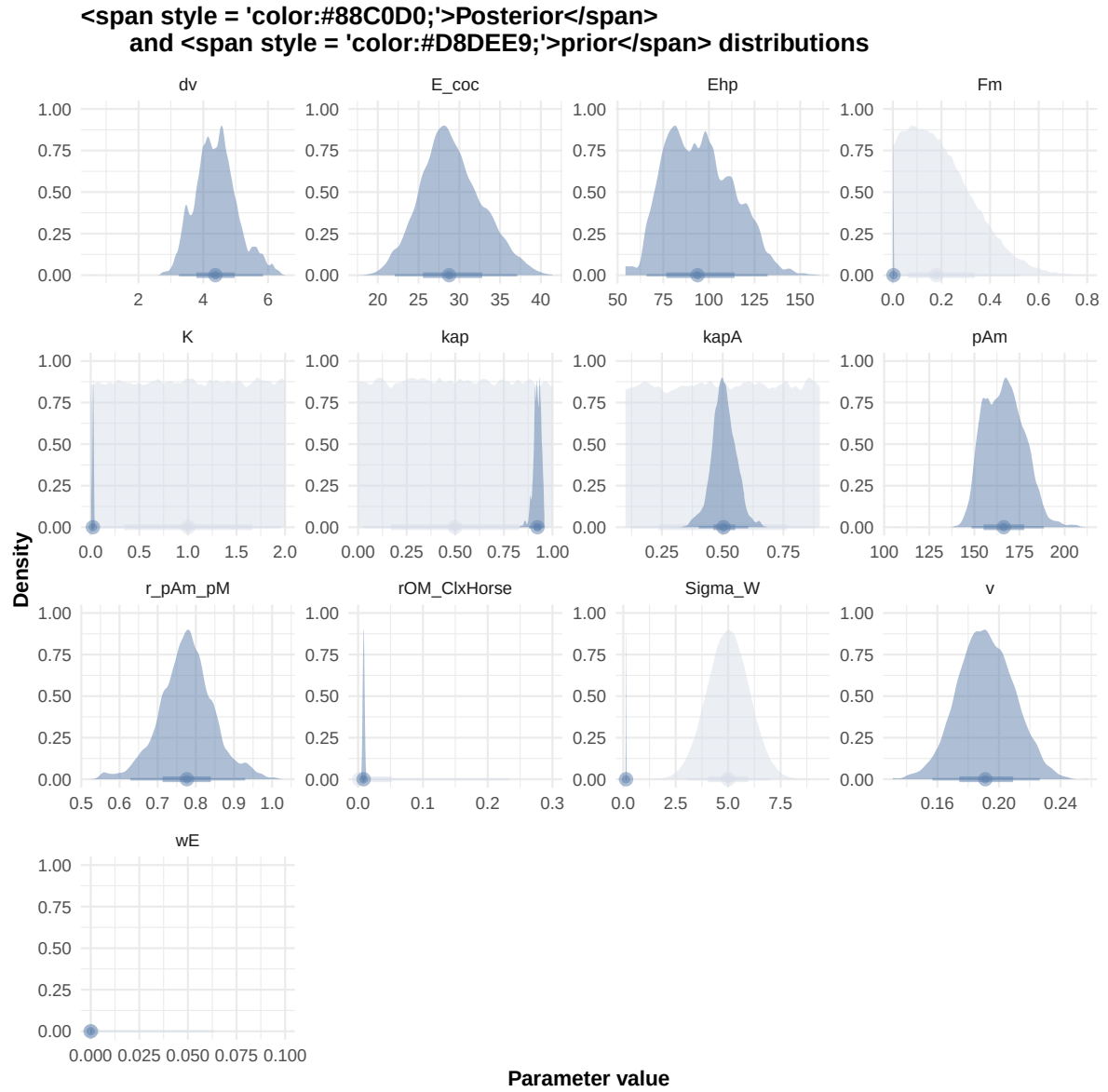


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

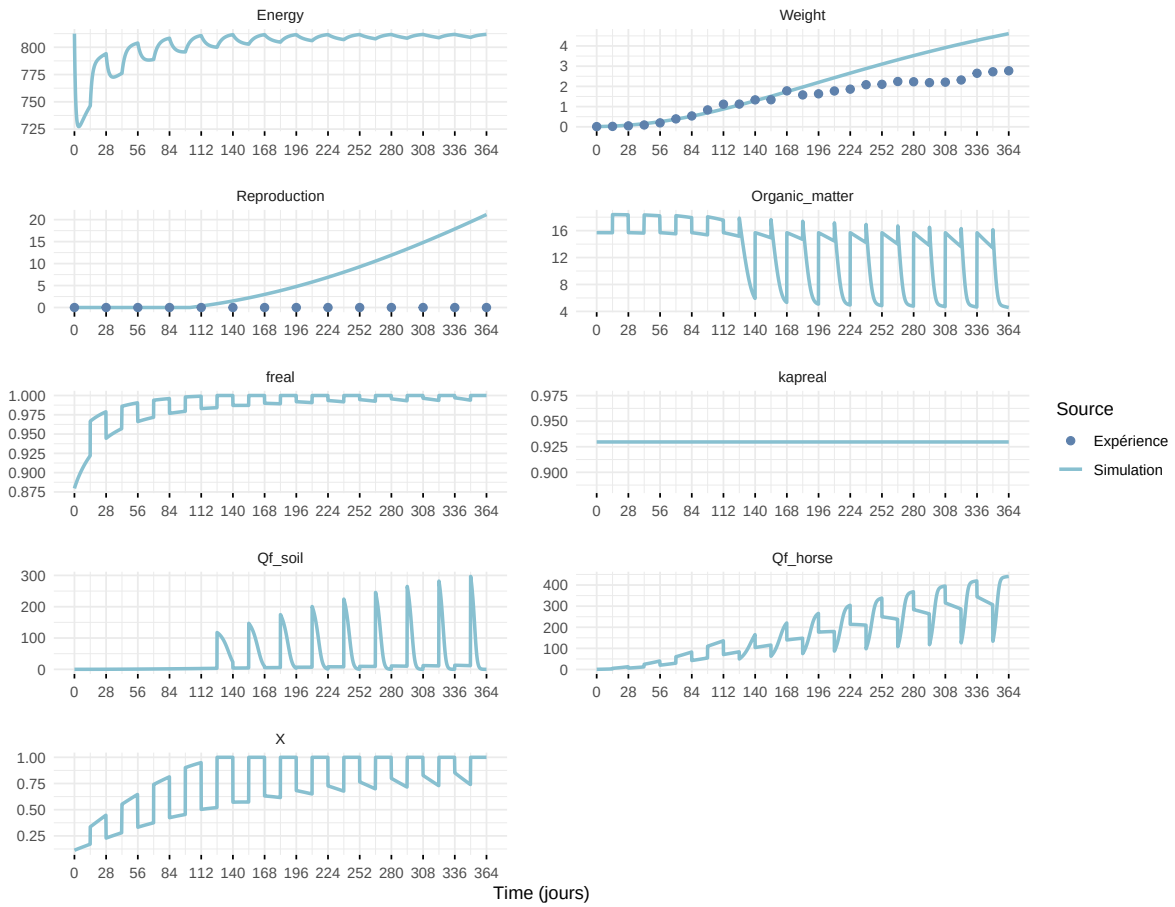
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexp
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



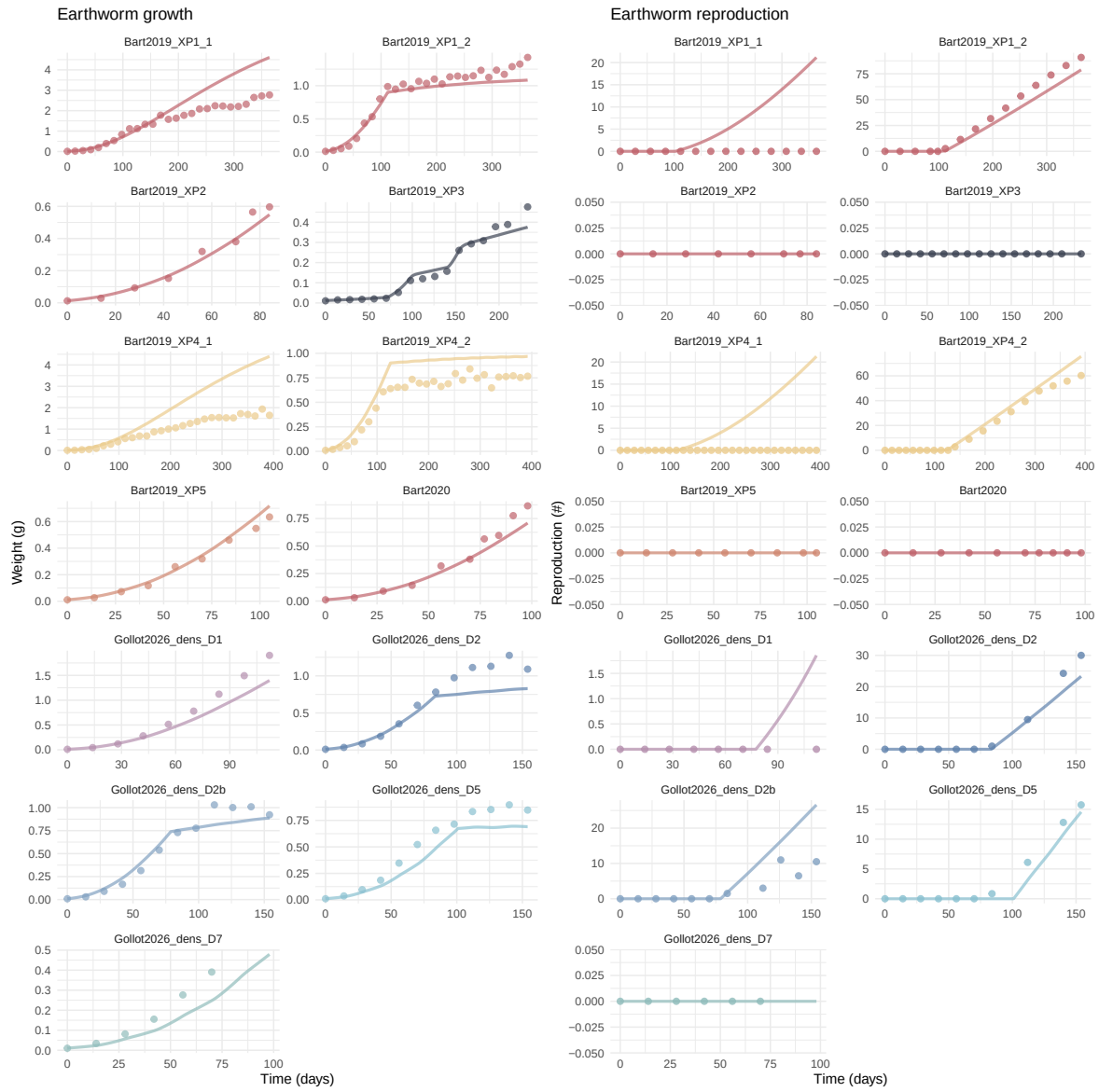
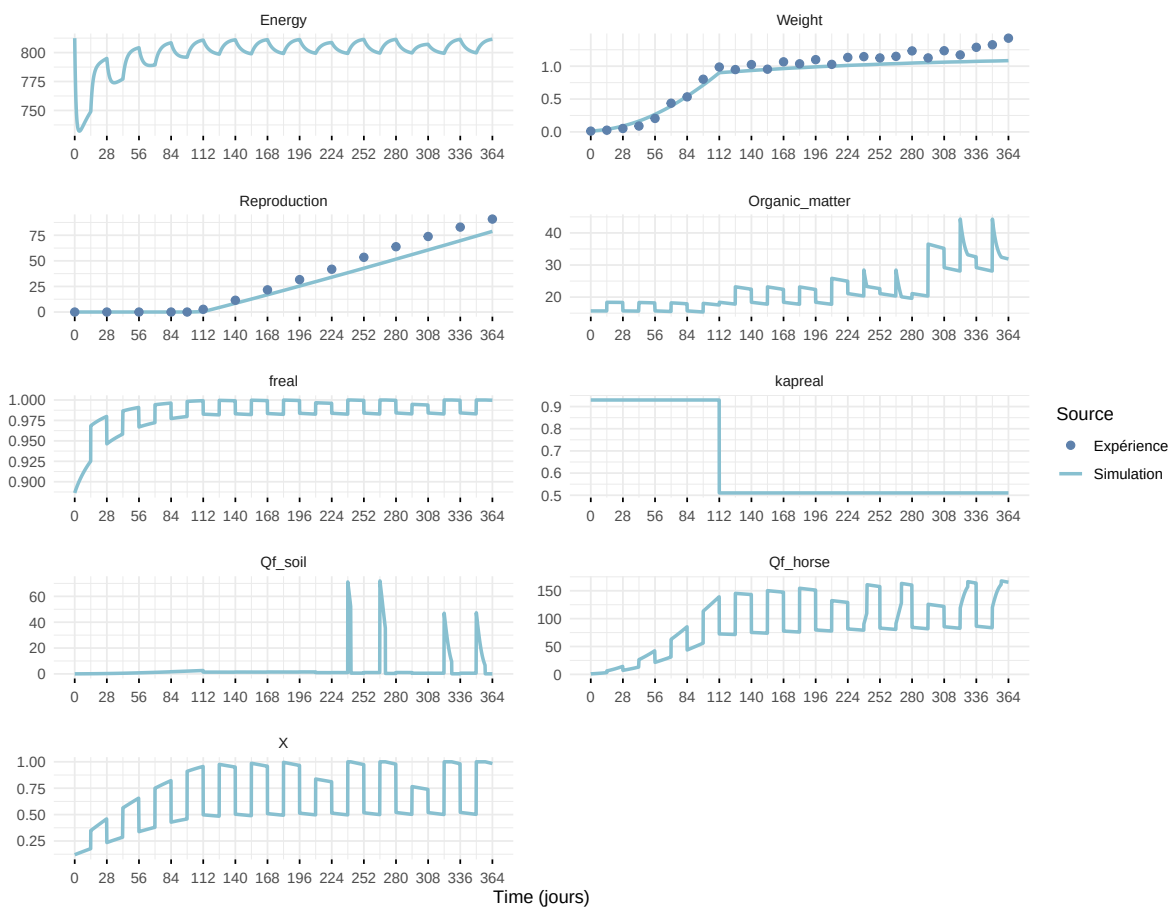
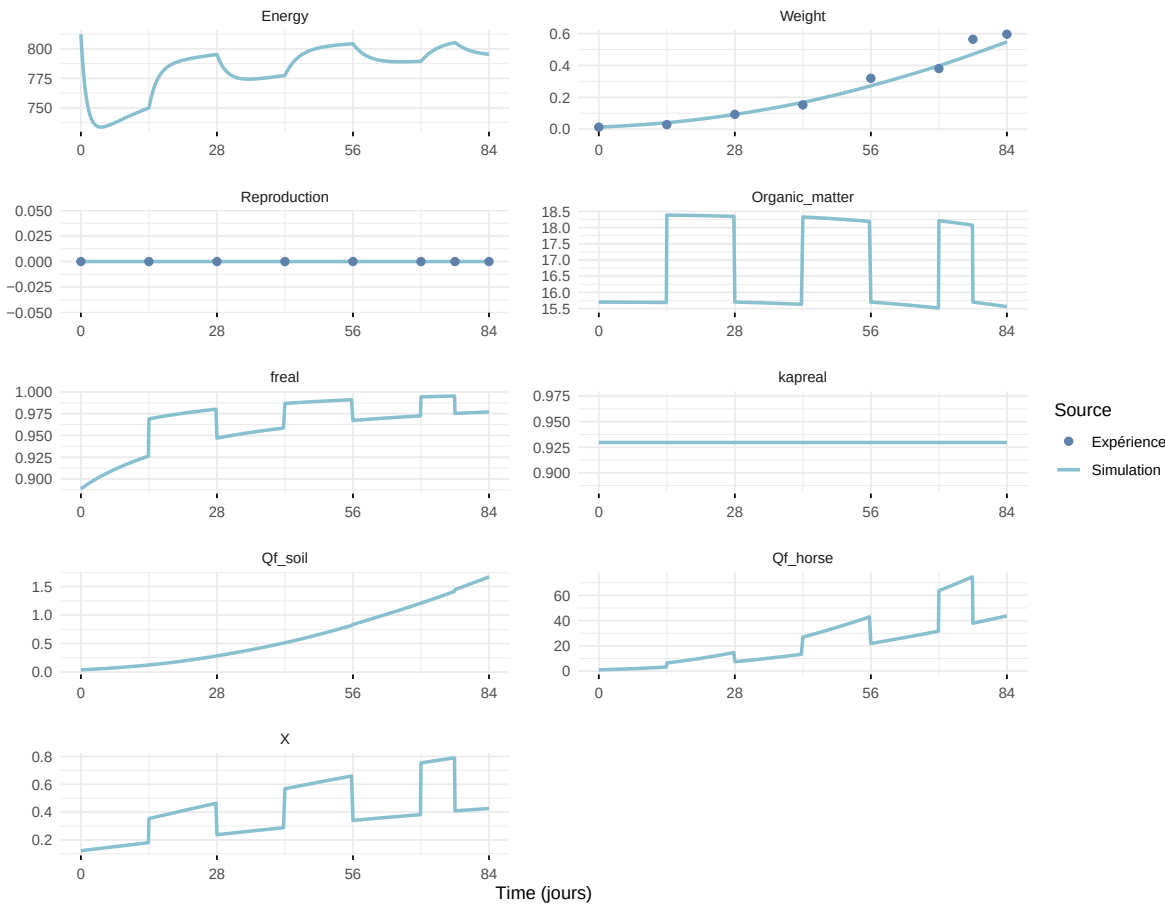


Figure 7: Simulations Weight and Reproduction.

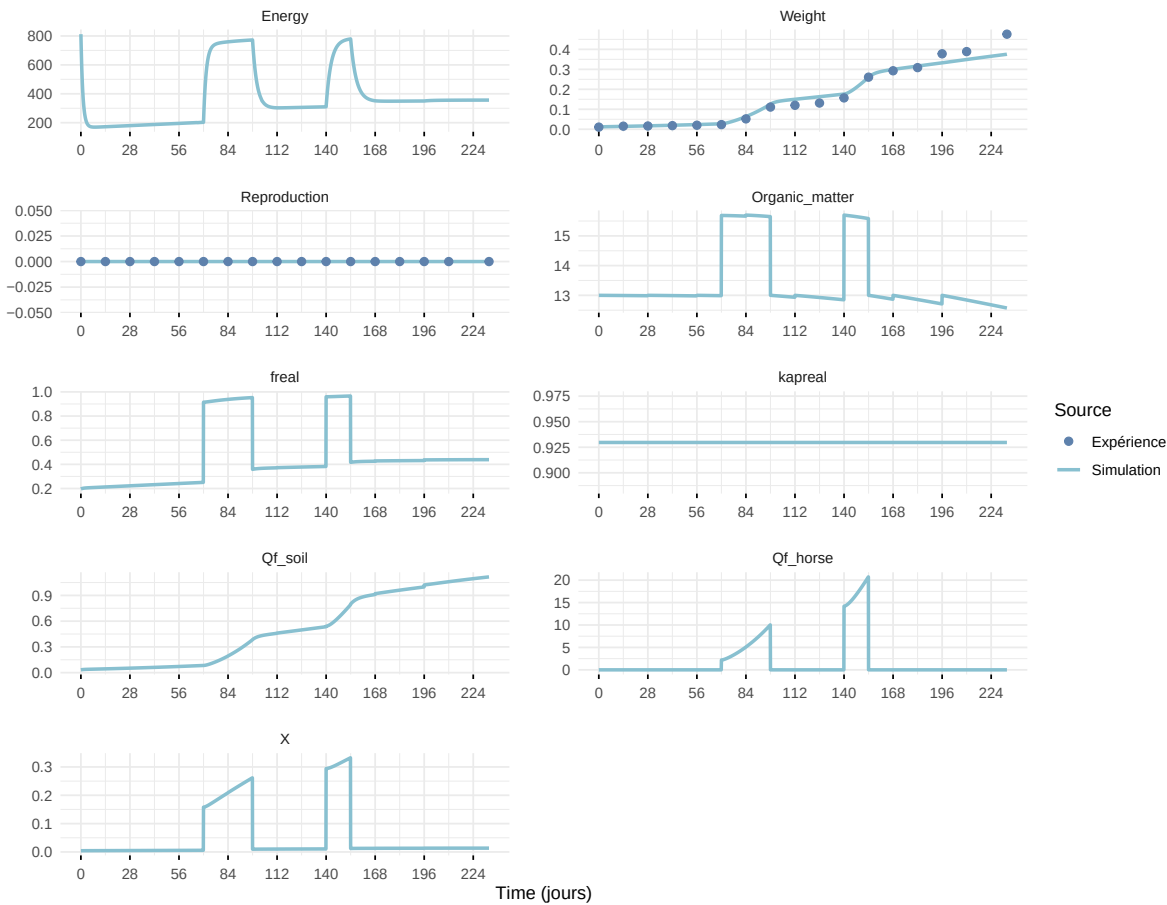
Expérience : Bart2019_XP1_2



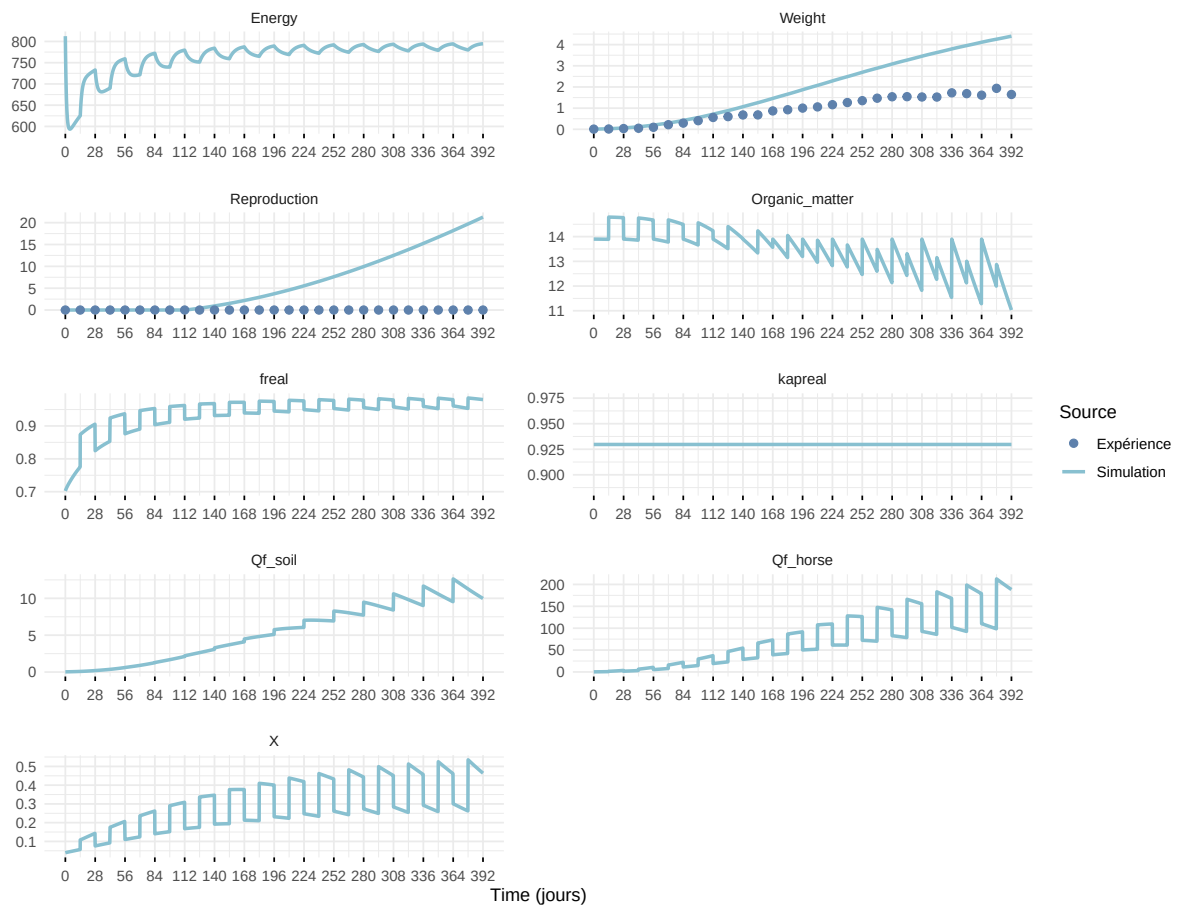
Expérience : Bart2019_XP2



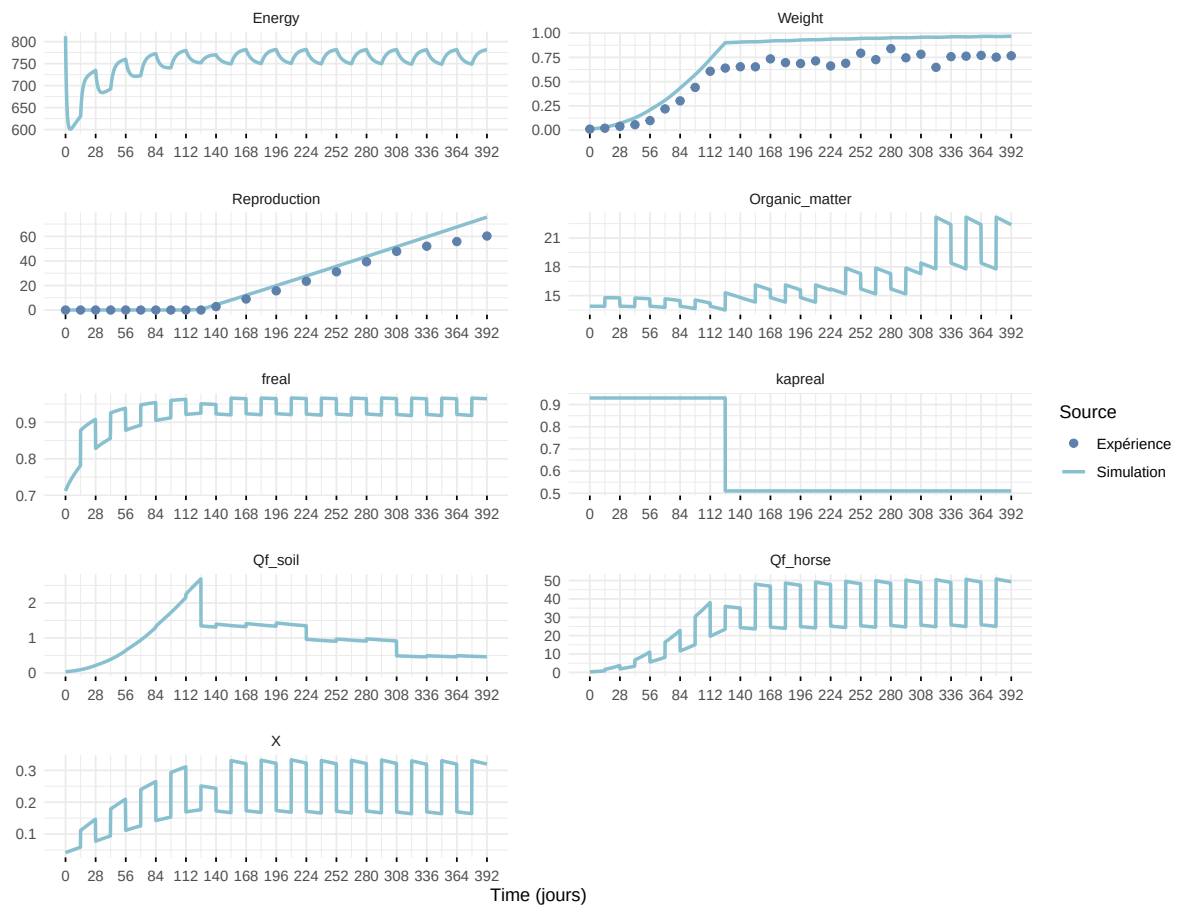
Expérience : Bart2019_XP3



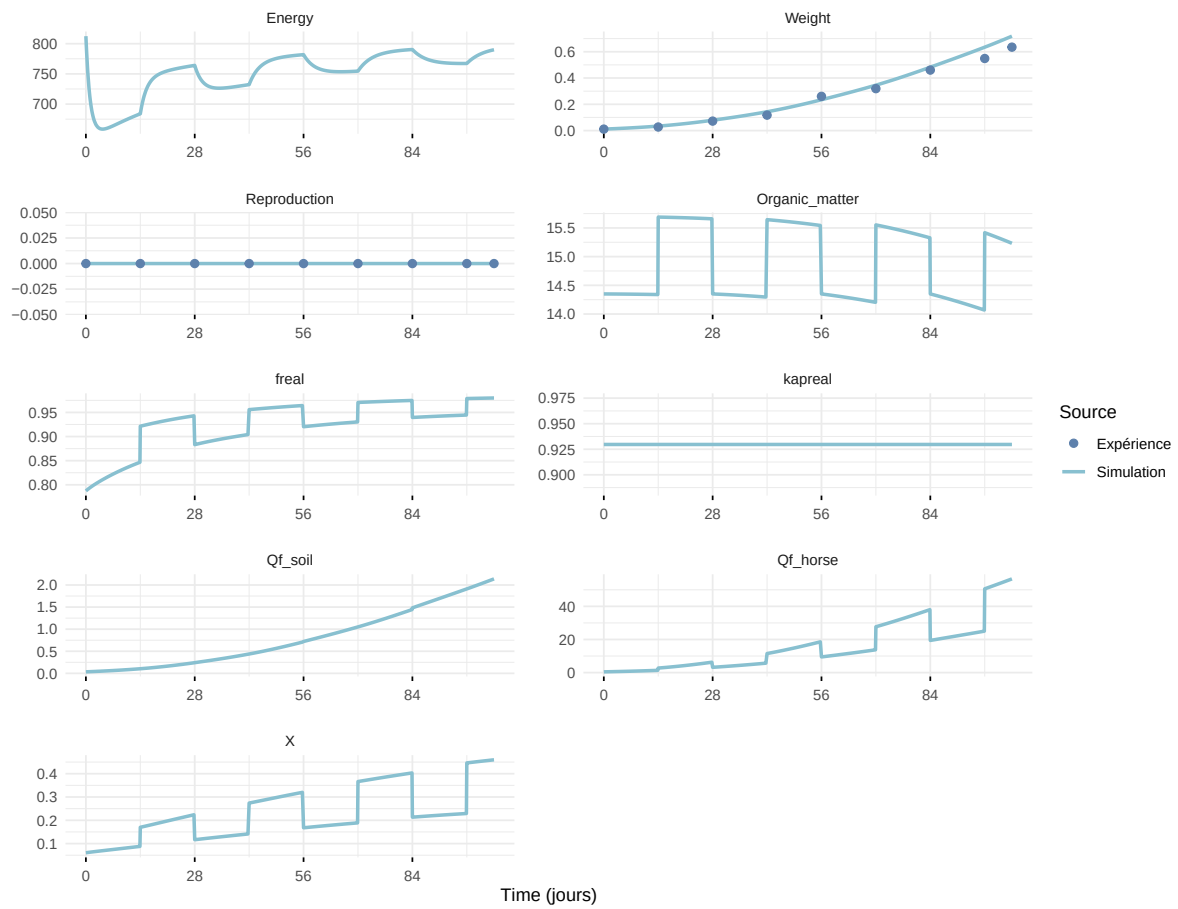
Expérience : Bart2019_XP4_1



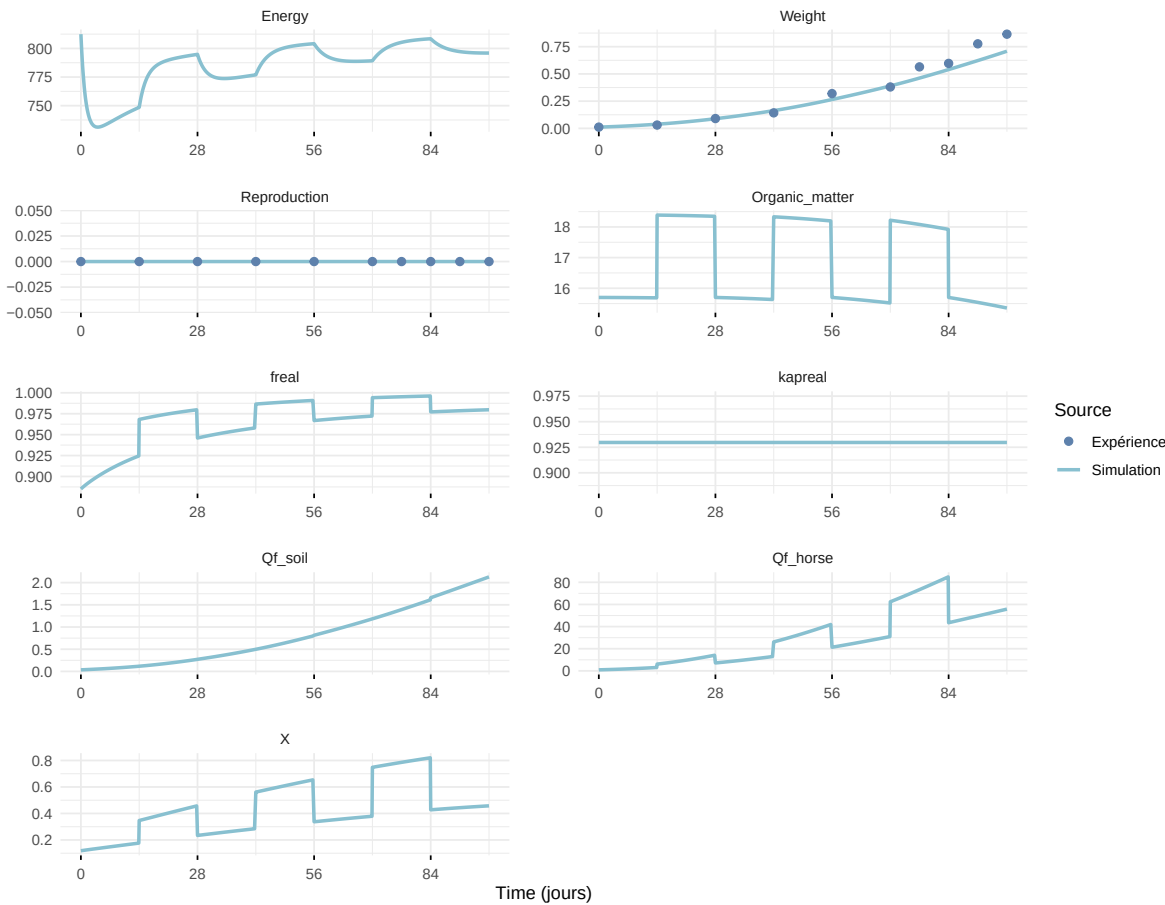
Expérience : Bart2019_XP4_2



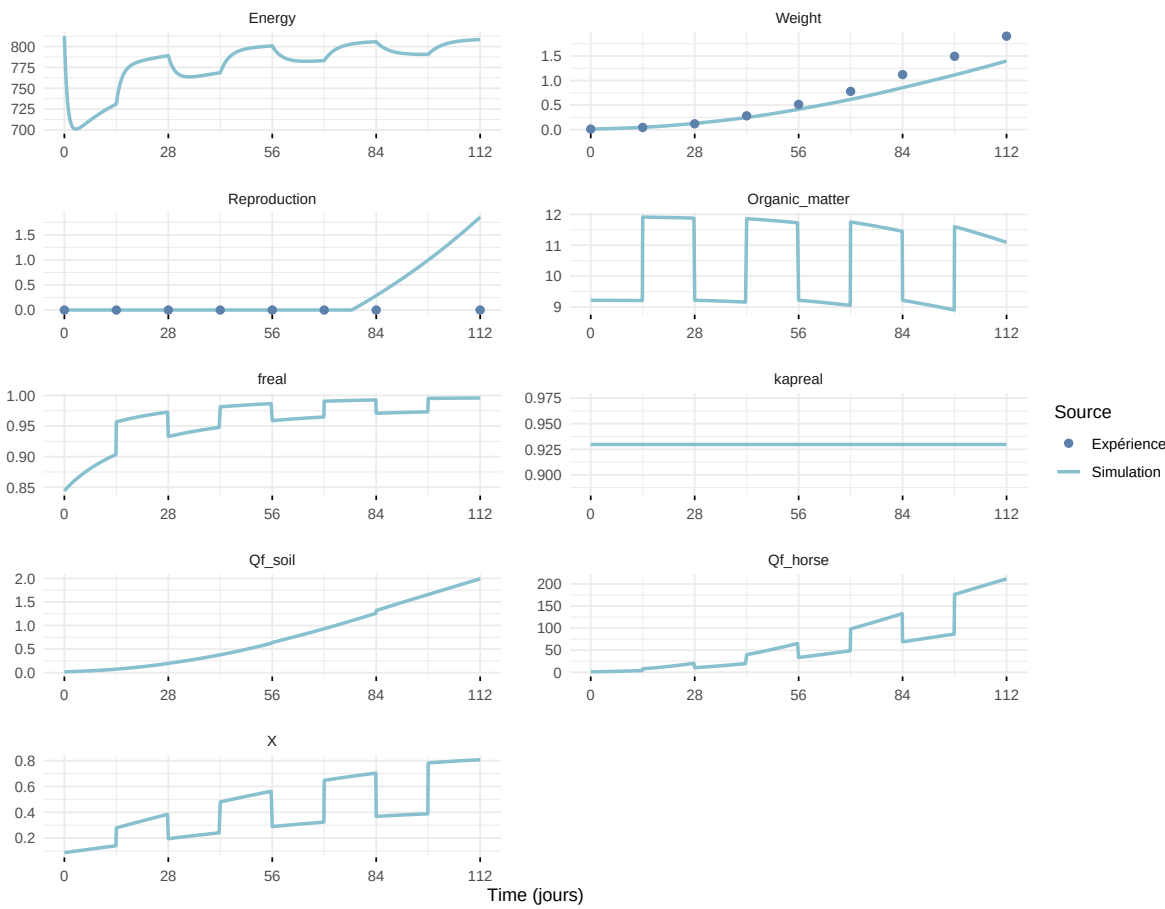
Expérience : Bart2019_XP5



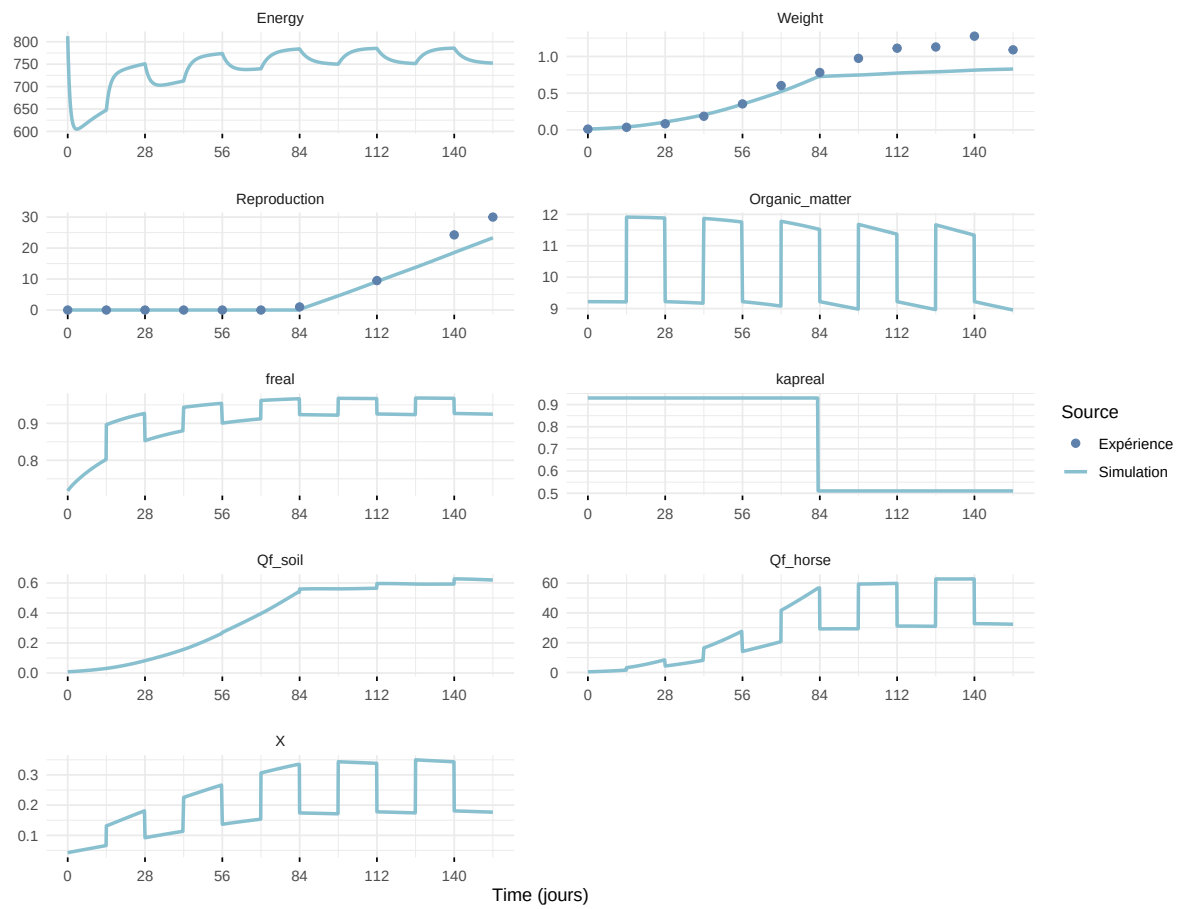
Expérience : Bart2020



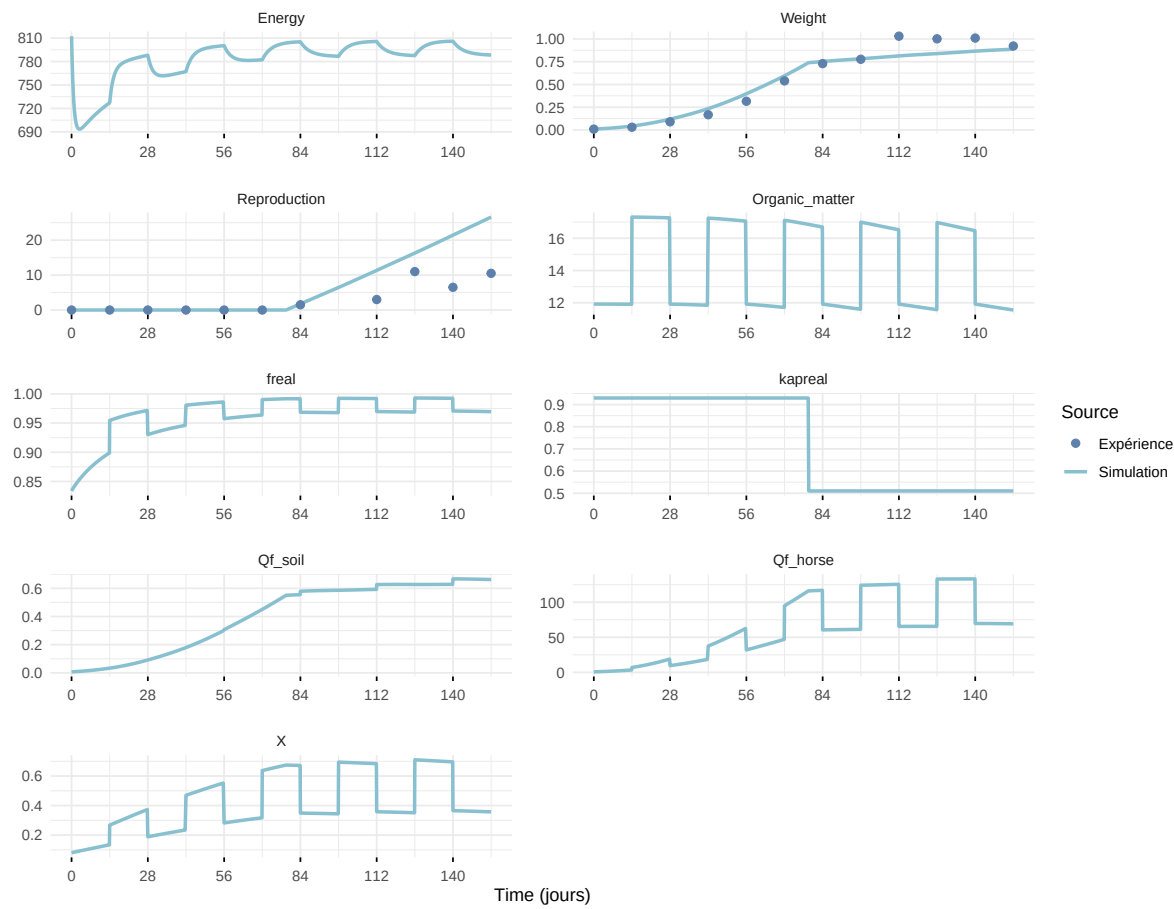
Expérience : Gollot2026_dens_D1



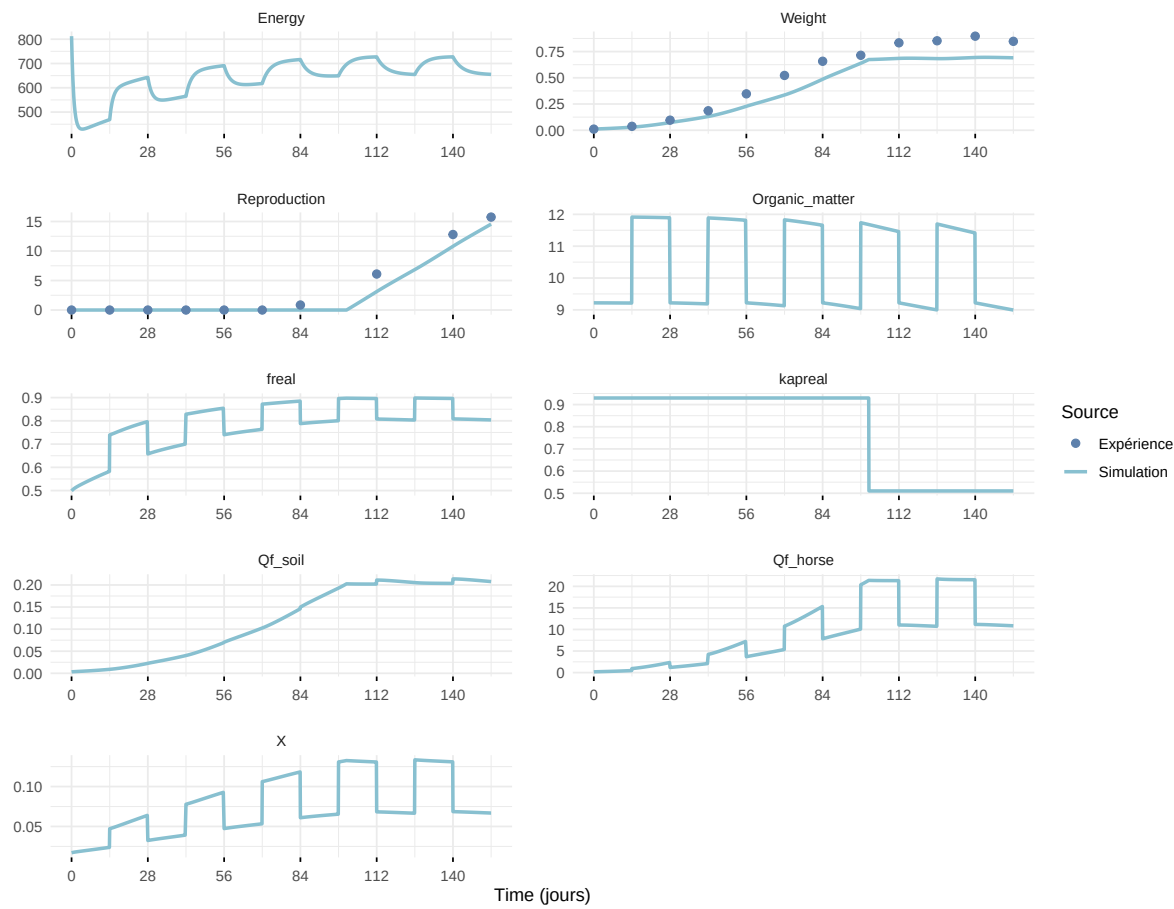
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

